Below Replacement, Unequally: Settlement-Era Regional Cultures and U.S. County Fertility

Author: Drew Patterson, independent researcher (ORCID 0009-0005-9551-4948)

DOI: 10.5281/zenodo.21396077 | **Replication:** https://github.com/Drew-Opexcell/american-nations-fertility

---

Abstract

The United States total fertility rate reached a record low of 1.60 in 2024, but the national figure conceals durable regional variation. This article maps contemporary fertility onto the American Nations model, a typology of settlement-era regional cultures (Woodard 2011) that follows county borders and has been applied to violence, health, and political outcomes, but not to fertility. Using CDC WONDER natality data (2021-2024 pooled), Census Bureau denominators, and a tract-level cultural-region crosswalk published by the Urban Institute, I compute total fertility rates for each nation and estimate county-level models with controls for religious adherence, income, education, and urbanicity. Nation-level TFRs range from 1.42 on the Left Coast and 1.54 in Yankeedom to 1.73 in Greater Appalachia and 1.77 in New France; every region is below replacement. Cultural region adds 11 percentage points of explained variance in county TFR beyond the controls, and the regional ordering echoes the geography of the antebellum fertility decline, in which Yankee-settled areas led the nation. Settlement-era cultural boundaries remain a live axis of American demographic variation two centuries later.

Keywords: fertility, regional culture, American Nations, demographic transition, religiosity, United States

---

1. Introduction

The U.S. total fertility rate fell from 2.12 in 2007 to 1.60 in 2024, the lowest level recorded (NCHS 2025). A substantial literature examines the decline's proximate drivers: the Great Recession, delayed marriage, housing and childcare costs, and changing preferences (Kearney, Levine, and Pardue 2022). Far less attention has gone to the geography of the decline. State fertility rates in 2024 span nearly a full child, from 1.27 in Vermont to 2.04 in South Dakota, a gap comparable to the difference between Canada and France.

Explanations for geographic variation typically invoke composition: some places are more religious, poorer, more rural, less educated. This article asks whether a deeper layer patterns the map: the regional cultures laid down by rival colonization streams before mass industrialization. The American Nations model (Woodard 2011), building on Fischer (1989) and the cultural geography of Zelinsky's "first effective settlement" doctrine, partitions North American counties into cultural regions defined by their dominant founding population: Puritan-settled Yankeedom, the Scots-Irish borderlands of Greater Appalachia, the plantation Deep South, Hispanic El Norte, and so on. The model has been used to analyze regional variation in violent deaths, life expectancy, and political behavior (e.g., Kottke et al. 2024), but to my knowledge no study has applied it to fertility.

Fertility is a natural test for the persistence of settlement cultures. Census-based child-woman ratios indicate that white fertility in the United States was falling from at least 1800, decades before industrialization or significant mortality decline and earlier than every developed nation except France (Haines and Hacker 2006), with deliberate marital fertility control detectable by 1830 (Hacker, Haines, and Jaremski 2021); Hacker (2003) offers a later reading of the onset, and the timing remains debated. The geography is less contested. Decline appeared first and fell fastest in New England and the Yankee-settled zones of upstate New York and the Western Reserve, diffusing outward along settlement lines. If founding cultures shaped the onset of fertility limitation in 1800, do their descendants' counties still differ in 2024, after controlling for the compositional factors that dominate modern accounts?

2. Data and methods

2.1 Cultural regions

County assignments come from the Urban Institute's National Center for Charitable Statistics census crosswalk, which codes every 2010-vintage census tract with its American Nations region. Because Woodard draws regional borders along county lines, aggregation is exact rather than approximate: no county in the file spans two nations, so every classified county maps to exactly one. Of 3,225 counties and county equivalents, 3,108 are classified and 117 are not, almost all of them in Puerto Rico, Alaska, and Hawaii.

Three features of the crosswalk constrain the analysis. First, it distinguishes ten regions rather than Woodard's eleven: New Netherland is not coded separately, and its territory is classified as Yankeedom. Because Yankeedom is this study's reference category, that merger deserves a direct test rather than a caveat. Reconstructing New Netherland from Woodard's stated boundaries, the five boroughs, the lower Hudson Valley, northern New Jersey, and western Long Island, yields a TFR of 1.539 against 1.511 for Yankeedom proper, a difference of 0.028 children, with New Netherland supplying 26 percent of the composite's exposure. The two nations are demographically near-identical, so the composite reference does not bias the gaps reported below. Two boundary points drive that result and are easy to get wrong: Suffolk County is eastern Long Island, which was New England-settled and belongs to Yankeedom, and Orange County (containing the Hasidic community of Kiryas Joel, TFR 2.24) lies above the lower Hudson. Including either, as any metropolitan-area proxy does, inflates New Netherland substantially and misstates the comparison. Southwestern Connecticut is New Netherland in Woodard's model but cannot be joined at county level (see below) and is excluded from both sides. Second, Alaska, Hawaii, and the territories are unclassified and excluded; First Nation and the Canadian extent of New France lie mostly outside U.S. county geography in any case. Third, Connecticut abolished its counties for statistical purposes after the crosswalk was built: vital statistics still report births under the legacy county codes while population estimates report the new planning regions, and the two do not nest. Because Connecticut lies entirely within Yankeedom, both sides aggregate correctly at the national level, but no county-level join is possible, so Connecticut is excluded from county-level models and from the identified-county sensitivity column.

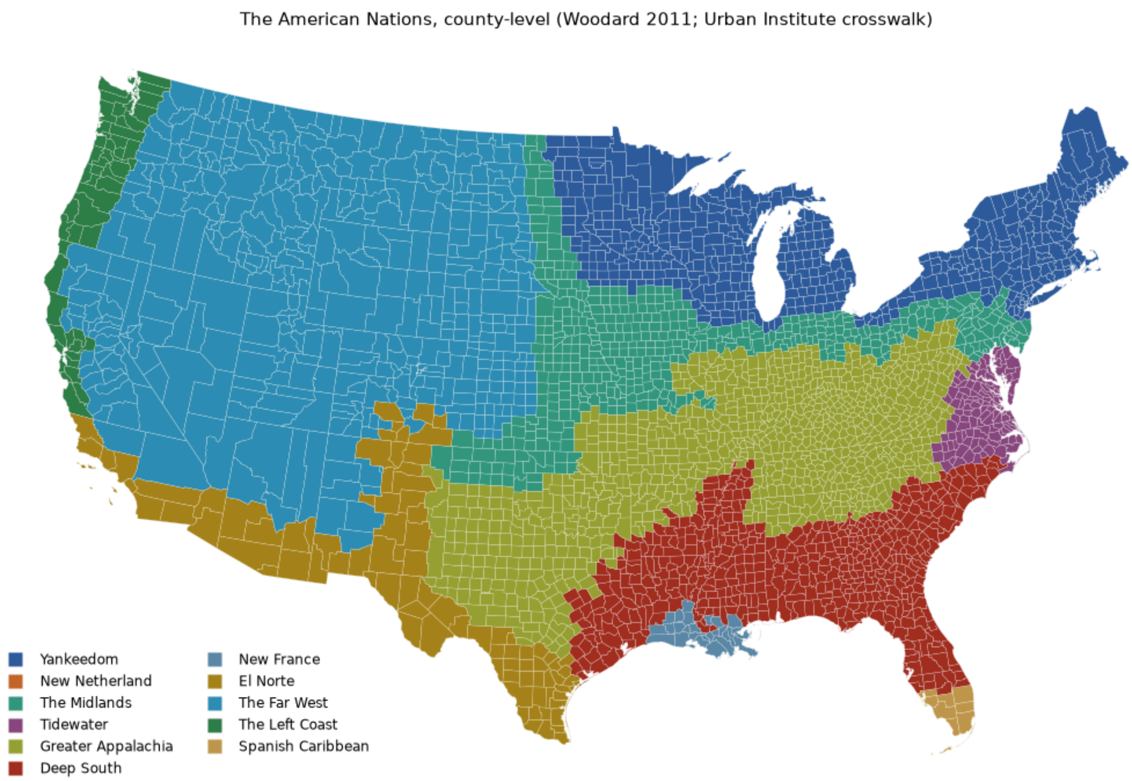

Figure 1. The American Nations, county level (Woodard 2011, via the Urban Institute NCCS crosswalk). Lower 48 shown; Alaska, Hawaii, and the territories are unclassified. New Netherland is not coded separately and appears here as Yankeedom.

2.2 Fertility measures

Births by county and five-year maternal age group, pooled over 2021-2024, come from CDC WONDER natality files. WONDER identifies counties with 2010 population of 100,000 or more, 578 of them, accounting for 79.5 percent of births in these years, and pools the remainder into a state-level residual. Of the 578, county-level models retain 566: Connecticut's eight legacy counties have no matching denominator (section 2.1), and four Alaska and Hawaii counties are unclassified. I compute age-specific rates using female population by five-year age group from the Census Bureau's Vintage 2024 county estimates, pooled over the same years as person-years of exposure. TFR is five times the sum of age-specific rates for ages 15-19 through 45-49, with births to mothers under 15 rated against women 10-14 and births at 50 and over folded into the 45-49 group, following NCHS convention.

Nation-level TFRs are computed two ways: (a) identified counties only, and (b) with each state's small-county residual births allocated to nations in proportion to the age-specific female population of that state's non-identified counties. The two series agree closely (maximum difference 0.06; Table 1), and no cell in the extract was suppressed. Allocation raises every nation's estimate, as expected given that small counties are rural and rural fertility is higher.

As a complete-coverage check that requires no allocation, I also compute the general fertility rate (births per 1,000 women 15-44) for every county from the Census Bureau's components-of-change series, which reports births for all counties without suppression. The GFR and TFR rankings agree.

2.3 County models

To test whether cultural region predicts fertility beyond composition, I estimate weighted least squares models of county fertility (GFR for all 3,098 modelable counties; TFR for the 566 identified counties) on nation indicators (Yankeedom as reference) and four controls: religious adherents per 1,000 population from the 2020 U.S. Religion Census (Grammich et al. 2023), log median household income (USDA ERS, 2022), percent of adults with a bachelor's degree or higher (ERS, 2019-2023), and urbanicity (metro indicator plus the 2023 Rural-Urban Continuum Code). Observations are weighted by female population 15-44; standard errors are clustered by state. The design is ecological by construction and is interpreted as such.

3. Results

3.1 Levels

Table 1 and Figures 1-2 present the core description. Every nation is below replacement. New France (1.77), the French-settled parishes of southern Louisiana, posts the highest TFR, followed by Greater Appalachia (1.73), the Midlands (1.70), and the Deep South (1.70).

New France warrants two caveats. First, it is not synonymous with Acadiana: the region includes Orleans and East Baton Rouge parishes, and its identified parishes range from Terrebonne (1.97) and Lafayette (1.90) in the Cajun southwest down to Orleans (1.40). The Cajun parishes carry the nation's position, at a general fertility rate of 66.0 against 54.8

nationally, while the remaining parishes sit near the national figure at 56.2. Second, New France is by some margin the smallest nation measured here, 26 parishes and roughly 37,000 births a year against Greater Appalachia's 936 counties and 714,000. Its rank should be read with that in mind, since smaller and more homogeneous aggregates sit further from the national mean by construction. The gap between New France and Greater Appalachia (0.035 children) is correspondingly slender, and the two also trade places by no more than that margin under the identified-county sensitivity in Table 1.

Yankeedom (1.54) and its Pacific offshoot the Left Coast (1.42) hold the bottom; the six states of Yankeedom's New England core sit lower still. El Norte (1.55), which carried the highest regional fertility in the country as recently as the mid-2000s, now sits below the national average, consistent with the post-2008 Hispanic fertility decline (Kearney, Levine, and Pardue 2022).

Table 1. Total fertility rate by American Nation, 2021-2024 pooled

Nation	TFR (with allocation)	TFR (identified counties)
New France	1.77	1.71
Greater Appalachia	1.73	1.68
The Midlands	1.70	1.64
Deep South	1.70	1.64
The Far West	1.67	1.64
Spanish Caribbean	1.65	1.64
Tidewater	1.59	1.56
El Norte	1.55	1.54
Yankeedom	1.54	1.52
The Left Coast	1.42	1.41

Yankeedom here includes New Netherland, whose own TFR is 1.539 against 1.511 for Yankeedom proper, so the composite is not a material distortion (see 2.1). The identified-

county column excludes Connecticut, which has no county-level denominator.

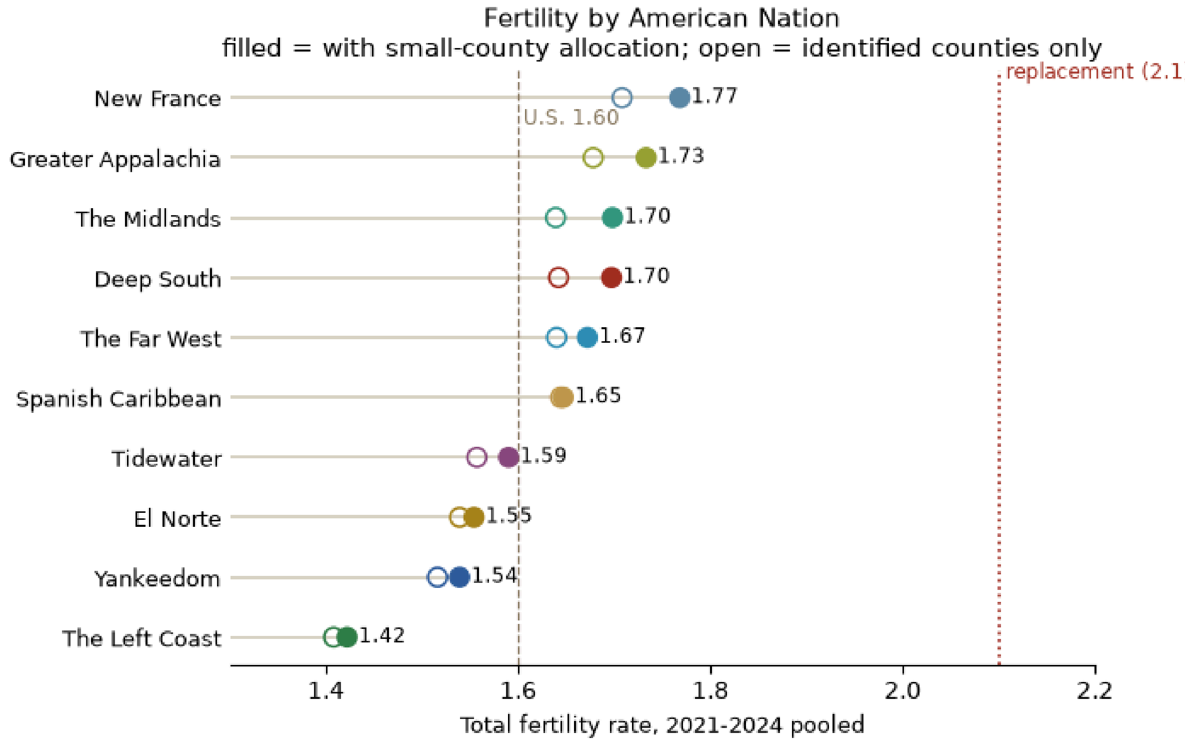

Figure 2. Total fertility rate by nation, 2021-2024 pooled. Filled markers include the small-county allocation; open markers use identified counties only. Every nation falls below replacement.

3.2 Culture net of composition

Table 2 reports the specification ladder for county TFR. Because regional culture plausibly shapes religiosity, education, and income themselves, controlled estimates should be read as the direct (residual) association and the unadjusted gaps as the total one; controlling for downstream channels can only understate culture's full imprint (the classic mediator problem). Nation indicators alone explain 12.3 percent of the weighted variance; controls alone explain 35.3 percent; together they reach 46.3 percent, so region adds 11.0 points over composition (3.9 points in the all-county GFR model).

Table 2. County TFR gaps vs. Yankeedom, 2021-2024 (identified counties, n = 566; state-clustered SEs)

Nation	Raw gap	Adjusted gap
New France	+0.19 (0.02)	+0.10 (0.03)
Greater Appalachia	+0.16 (0.03)	+0.13 (0.02)
The Far West	+0.12 (0.07)	+0.00 (0.03)
Deep South	+0.12 (0.06)	+0.06 (0.04)
The Midlands	+0.12 (0.05)	+0.11 (0.04)
Spanish Caribbean	+0.11 (0.02)	+0.10 (0.02)
Tidewater	+0.04 (0.04)	+0.10 (0.03)
El Norte	+0.00 (0.09)	-0.09 (0.09)
The Left Coast	-0.12 (0.03)	-0.11 (0.03)

Adjusted: net of religious adherence, log median income, percent BA+, and urbanicity. Figure 3 plots the adjusted column.

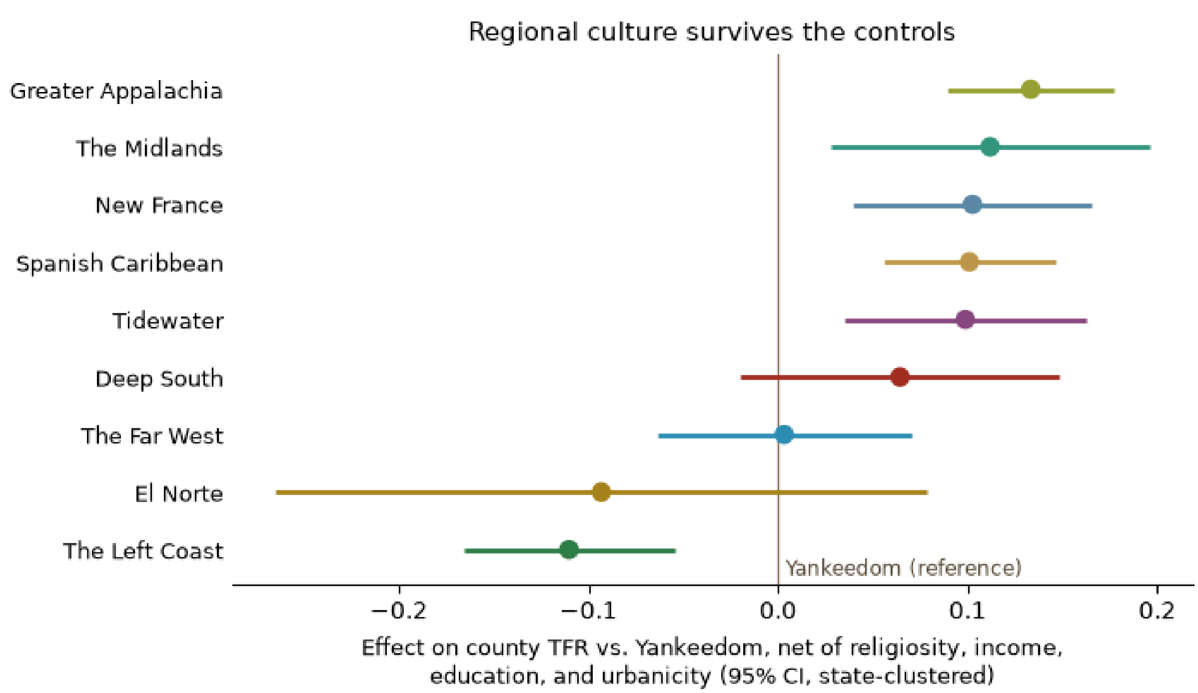

Figure 3. Nation effects on county TFR relative to Yankeedom, net of religious adherence, income, education, and urbanicity. Bars are 95 percent confidence intervals with state-

clustered standard errors.

The comparison is itself informative. Greater Appalachia, the Midlands, the Spanish Caribbean, and the Left Coast keep essentially their whole gap after adjustment: their distinctiveness is not a composition story. The Far West's raw advantage, by contrast, is fully absorbed by the controls (chiefly religiosity and rurality), and about half of New France's is. Tidewater is a case of suppression: its raw gap is small because the region is unusually affluent and educated, and the cultural difference emerges only once those are held constant. Religious adherence retains a strong independent association (+0.06 children per additional 100 adherents per 1,000 population); education is negatively and income positively associated with county fertility within regions.

Two features of the pattern deserve emphasis. First, the net cultural spread is substantively large: roughly a quarter of a child separates Greater Appalachia from the Left Coast after composition is held constant, in a country whose entire 2007-2024 fertility decline was about half a child. Second, the ordering is historically legible. The regions that led the antebellum fertility transition, Yankeedom and its Left Coast offshoot, occupy the floor of the contemporary distribution; the borderland and plantation south regions that adopted fertility limitation latest in the nineteenth century remain the most fertile today, with or without controls.

4. Discussion

The American Nations map was drawn from colonial settlement history, not from modern demographic data. That a typology fixed on seventeenth- and eighteenth-century founding populations still sorts county fertility in 2021-2024, within levels of religiosity, income, education, and urbanicity, suggests that regional culture operates as an independent axis of demographic variation rather than a shorthand for composition. This is consistent with work in the second-demographic-transition tradition showing spatially patterned family behavior in the United States that tracks cultural rather than purely economic geography (Lesthaeghe and Neidert 2006), and it extends the growing American Nations health literature to fertility.

The historical continuity is striking. Census reconstructions place the onset of sustained fertility decline in the white population by 1800, earliest in New England and the Yankee settlement stream (Haines and Hacker 2006). Two centuries later, those same regions anchor the bottom of the distribution: Yankeedom posts the second-lowest TFR of any nation, and the Left Coast, which Woodard describes as a Yankee settlement founded by missionaries and merchants, posts the lowest and is the only nation significantly below the Yankee reference in every specification. Whatever bundle of norms produced the world's second-earliest fertility transition appears, in attenuated form, still to distinguish these regions.

The residual regional differences reflect some mix of intergenerational transmission and sorting that this design cannot decompose. Two of the candidate channels transmit across generations, and gene-culture coevolution entangles them (Richerson and Boyd 2005). The first is socialization: regional norms about family size pass from parents to children. The second is genetic: reproductive behavior is partly heritable (Kohler, Rodgers, and Christensen 1999; Barban et al. 2016), founding populations differed in ancestry, and polygenic scores cluster geographically through migration-driven sorting (Abdellaoui, Hugh-Jones, and Yengo 2019). Testing the genetic channel would require mean polygenic scores by region, which do not exist for these populations. Two further explanations involve no transmission at all: selective migration (low-fertility-preference individuals sorting into Left Coast and Yankee metros) and unmeasured composition (dimensions of religiosity or opportunity the controls miss; adherence counts congregational supply, not belief). Estimates should be read as descriptive of place, not causal for individuals. The ecological design is the study's main limitation; period TFR's sensitivity to tempo effects is another, though tempo distortion would have to vary by nation to manufacture the cross-sectional pattern.

Two regional results merit their own follow-up. El Norte's fall from the country's fertility engine to below-average illustrates how quickly a regional regime can change when immigrant generational composition shifts; its coefficient is already indistinguishable from Yankeedom's. And New France's position at the top of the U.S. table inverts the trajectory of its sibling society in Quebec, which moved from world-record marital fertility to a record-low 1.33 in 2024 (Institut de la statistique du Quebec 2025), below every American nation. Two branches of one founding population, separated by a border, now sit at opposite ends of the distribution:

a reminder that founding stock is not destiny, and that institutions and subsequent history do heavy lifting.

5. Conclusion

Every American nation is now below replacement, but they are not below it equally. County fertility in 2021-2024 still bears the imprint of the settlement-era cultural map: about a quarter-child of net regional spread that religiosity, income, education, and urbanicity do not account for, and an ordering that recapitulates the geography of the nineteenth-century fertility transition.

The two findings point in opposite directions, and both are true. In 1800, New England was limiting births while the rest of the country averaged near seven children; today the national rate is 1.60 and Yankeedom's is 1.54. On the level, the country has converged on the regime its earliest transitioning region pioneered. On the ordering, the map that region belonged to has not dissolved at all. Individual-level data with regional identifiers would be the natural next test of whether these are effects of place, selection into place, or both.

Data availability

All inputs are public: CDC WONDER natality files, Census Bureau Vintage 2024 county population estimates, the 2020 U.S. Religion Census county file (ARDA/OSF), USDA ERS county education and income files, and the Urban Institute NCCS geographic crosswalk. The complete analysis pipeline (Python, no proprietary dependencies), derived datasets, and figures are archived at https://doi.org/10.5281/zenodo.21396077 and maintained at https://github.com/Drew-Opexcell/american-nations-fertility.

References

- Abdellaoui, Abdel, David Hugh-Jones, and Loic Yengo. 2019. "Genetic Correlates of Social Stratification in Great Britain." *Nature Human Behaviour* 3(12): 1332-1342.

- Barban, Nicola, Rick Jansen, Ronald de Vlaming, et al. 2016. "Genome-wide Analysis Identifies 12 Loci Influencing Human Reproductive Behavior." *Nature Genetics* 48(12): 1462-1472.
- Fischer, David Hackett. 1989. *Albion's Seed: Four British Folkways in America.* New York: Oxford University Press.
- Grammich, Clifford, Erica Dollhopf, Mary Gautier, Richard Houseal, Dale E. Jones, Alexei Krindatch, Richie Stanley, and Scott Thumma. 2023. *2020 U.S. Religion Census: Religious Congregations and Membership Study.* Association of Statisticians of American Religious Bodies.
- Hacker, J. David. 2003. "Rethinking the 'Early' Decline of Marital Fertility in the United States." *Demography* 40(4): 605-620.
- Hacker, J. David, Michael R. Haines, and Matthew Jaremski. 2021. "Early Fertility Decline in the United States: Tests of Alternative Hypotheses Using New IPUMS Complete-Count Census Microdata and Enhanced County-Level Data." *Research in Economic History* 37: 89-128.
- Haines, Michael R., and J. David Hacker. 2006. "The Puzzle of the Antebellum Fertility Decline in the United States: New Evidence and Reconsideration." NBER Working Paper 12571.
- Institut de la statistique du Quebec. 2025. *Bilan demographique du Quebec, edition 2025.* Quebec City: Institut de la statistique du Quebec.
- Kearney, Melissa S., Phillip B. Levine, and Luke Pardue. 2022. "The Puzzle of Falling US Birth Rates since the Great Recession." *Journal of Economic Perspectives* 36(1): 151-176.
- Kohler, Hans-Peter, Joseph L. Rodgers, and Kaare Christensen. 1999. "Is Fertility Behavior in Our Genes? Findings from a Danish Twin Study." *Population and Development Review* 25(2): 253-288.
- Kottke, Thomas E., Nicolaas P. Pronk, Colin Woodard, and Ross Arena. 2024. "The Potential Influence of Firearm Violence on Physical Activity in the United States." *The American Journal of Medicine* 137(5): 426-432.

- Lesthaeghe, Ron J., and Lisa Neidert. 2006. "The Second Demographic Transition in the United States: Exception or Textbook Example?" *Population and Development Review* 32(4): 669-698.
- National Center for Health Statistics. 2025. "Births in the United States, 2024." NCHS Data Brief 535. Hyattsville, MD.
- Richerson, Peter J., and Robert Boyd. 2005. *Not by Genes Alone: How Culture Transformed Human Evolution.* Chicago: University of Chicago Press.
- U.S. Department of Agriculture, Economic Research Service. 2024. *County-Level Data Sets: Education and Unemployment.* Washington, DC.
- Woodard, Colin. 2011. *American Nations: A History of the Eleven Rival Regional Cultures of North America.* New York: Viking.